



STRUCTURED NOTES

CAPTURE • ORGANIZE • UNDERSTAND • APPLY

Name Samyl Jimenez
Course / Subject Proc. P. Mec.
Speaker / Class _____
Date & Time _____

KEYWORDS & KEY CONCEPTS

- Nodo
- Lado
- Grafo
- Bucle
- lados Paralelos



Focus on the key concepts that generate value and understanding.

ACTIONS / NEXT STEPS

- ☐
- ☐
- ☐
- ☐
- ☐

Turn knowledge into action.
Small steps, big results.

NOTES

Title: Capitulo 7
Topic: Grafos.

Nace en 1735 con Leonhard Euler al resolver el problema de los Puentes de Königsberg. Demostro que para recorrer un grafo pasando por cada Arista una sola vez, los Vertices de entrada y salida deben tener Valencia Par.

un grafo
 $G = (V, E)$ consta de un Conjunto no Vacío de vertices V y un Conjunto de lados o Aristas E .

QUESTIONS & REFLECTIONS



What questions do I have?



What is unclear or confusing?



What ideas connect?



How can I apply this?



What assumptions should I challenge?



SUMMARY

Write a brief summary of the main points.



KEY TAKEAWAY

Capture the most important insight.

Carlos Richarcho Viñue

LIFELONG LEARNING

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Name Samuyl Jimenez
Course / Subject ppmt
Speaker / Class _____
Date & Time _____

KEYWORDS & KEY CONCEPTS

- Grafo simple
- Digrafo
- Grafo ponderado
- Grafo Bipartido
- Grafo Completo



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NOTES

Title:

Capitulo 7

Topic:

Tipos de Grafos

Grafos no dirigidos vs Dirigidos: En los dirigidos, las relaciones son unidireccionales. En los no dirigidos son bidireccionales.

Grafo Bipartido Completo: Todos los vertices del grupo A se conectan con todos los del grupo B.

Multigrafos: Grafos que permiten tener aristas paralelas conectando los mismos nodos.

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Name Samyl Jimenez
Course / Subject ppmt
Speaker / Class _____
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KEYWORDS & KEY CONCEPTS

- Matriz de Adyacencia
- Matriz de Incidencia



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ACTIONS / NEXT STEPS

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- ☐
- ☐
- ☐
- ☐



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NOTES

Title: Capítulo 7

Topic: Representación matricial de Grafos

Construcción de matriz adyacente:
si el vertice U_i está conectado a V_j se coloca 1 en la posición (j, i) de la matriz; de lo contrario se coloca un 0.

Construcción de la matriz incidencia:
las filas representan vertices y las columnas Aristas.

Se coloca un 1 en la intersección si la arista incide al vertice



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Name Samuel Jimenez
Course / Subject Apptm
Speaker / Class _____
Date & Time _____

KEYWORDS & KEY CONCEPTS

- Camino
- Circuito
- Euleriano
- Hamiltoniano

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ACTIONS / NEXT STEPS

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Title: Capitulo 7
Topic: Caminos y Circuitos

Teorema de Euler:
un grafo conexo tiene un camino de Euler si y solo si todos sus vertices tienen valencia par.

Un grafo conexo tiene un camino si y solo si tiene exactamente 2 vertices de valencia impar

Longitud de un camino:
Es el numero de Aristas que los componen.

QUESTIONS & REFLECTIONS

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Name Samyl Jimenez
Course / Subject Dptm
Speaker / Class _____
Date & Time _____

KEYWORDS & KEY CONCEPTS

- Isomorfismo
- Invariante
- Direccion
-
-
-
-
-
-



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ACTIONS / NEXT STEPS

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NOTES

Title: Capitulo 7

Topic: Isomorfismo de grafos.

Como descartar con isomorfismo rapidamente: si dos grafos tienen diferencias en cualquiera de las siguientes propiedades invariantes, no son isomorfos

- Numero de Vertices

- Numero de Aristas

- Conjunto o Valencia de Secuencia de los Vertices

- Conectividad y Cantidad de Subcircuitos

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Name Samir Jyoti
Course / Subject pptm
Speaker / Class _____
Date & Time _____

 Date & Time

 KEYWORDS & KEY CONCEPTS

Arbol
Bais
Wagas
Altura
Camino unico



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NOTES

Title: Capítulo 8
Topic: Introducción y Propiedades de los Árboles

Propiedades fundamentales de un árbol de n vértices

1. Es acíclico

2. Es Conexo

3. El numero de Aristas o lados es 0

4- Si se añade una sola arista nueva, se creara obligatoriamente un ciclo

5-Si se elimina una arista, el árbol dejara de ser Conexo.

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Name Samy Jaramel
Course / Subject Optm.
Speaker / Class _____
Date & Time _____

KEYWORDS & KEY CONCEPTS

- Arbol binario
- Arbol binario
- Completo
- Bosque
- Nodos Internos

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NOTES

Title: Capítulo 8
Topic: Tipos de Arboles y Bosques

Propiedades de un arbol binario
Completo con i nodos internos:

Numero de Hojas = $i + 1$

Numero Total de Vertices (V_T) =
 $2i + 1$.

Bosques: Estructuralmente, si un arbol
de n nodos los removemos
algunas de sus aristas clave
sus desconectar los subarboles
individuales

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Name Samy Jomery
Course / Subject Python
Speaker / Class _____
Date & Time _____

KEYWORDS & KEY CONCEPTS

- Arbol generador
- MST
- Algoritmo de Prim
- Algoritmo de Kruskal
-
-
-



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NOTES

Title: Capitulo 8
Topic: Arboles generales y de peso minimo

Diferencias entre Prim y Kruskal

Prim: Comienza en un nodo raíz específico y crece hacia afuera paso a paso de forma conexa.

Kruskal: No requiere que el subgrafo intermedio este conectado. Agrega aristas baratas de cualquier lugar del grafo, unificando fragmentos del bosque hasta conformar el arbol final.

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Name Sarayli Jimenez
Course / Subject _____
Speaker / Class _____
Date & Time _____

KEYWORDS & KEY CONCEPTS

- Arbol de expresion
- Recorrido Preorden
- Recorrido Inorden
- Recorrido Post
- Orden

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NOTES

Title: Capitulo 8
Topic: Recorrido y expresion de Arboles

Los Tres Métodos de lectura:

Preorden: Utilizado habitualmente para clonar estructuras de Arboles o generar código ejecutable

Inorden: Produce la expresión matemática en su formato tradicional humano (info).

Postorden: Genera notación polaca inversa, usada por compiladores y calculadoras científicas porque no requiere parentesis para resolver operaciones.

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